INCH-POUND

A–A–59590/12A
19 January 2006
SUPERSEDING
A-A-59590/12
5 April 2002

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

HOLDER, ELECTRICAL CARD, WEDGE RETAINERS, 3 PIECE, SCREW ACTUATED DRIVE, .240 X .180 INCH BODY SIZE, WITH SCREW RETENTION

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring electrical card holders described herein shall consist of this document and the latest issue in effect of A–A–59590.

CLASSIFICATION/PART IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES).

AA59590	/12	*	*	* *	W, D, E
CID number	Applicable CID specification sheet	Finish	Mounting	Length (see table I)	Options

Example: AA59590/12CT48 is the PIN for a chemical film finished, 4.8 inch long card holder with a lock nut on the screw for captivation. The card holder also features two tapped mounting holes for use with 0–80 UNF fasteners.

SALIENT CHARACTERISTICS.

Performance. Card holders shall hold the circuit card assembly firmly in its installed position. When the card holder is installed properly, it is capable of withstanding 60g/6ms of shock, 25 G-rms of vibration, and provides from 2 to 4 degrees C/W/inch thermal resistance transfer between the circuit card assembly and the heat sink surfaces.

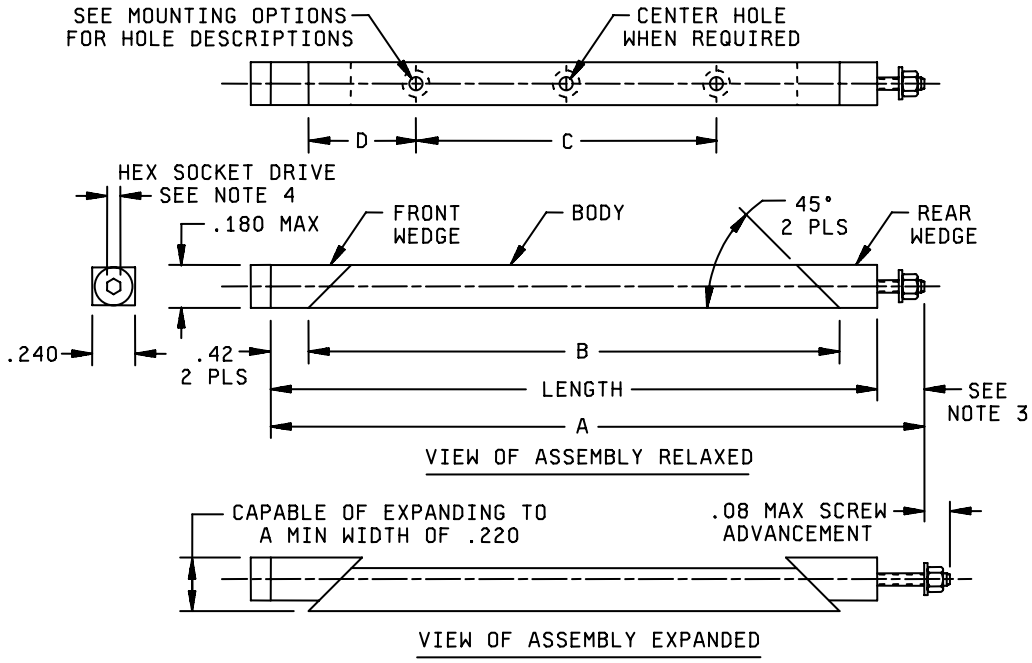
Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified on figures 1, 2, 3, 4, in table I, and A–A–59590.

Screw retention (locknut). The screw retention feature intended to prevent unintentional disassembly of the card holder shall be as specified in figure 2. The locknut shall be fabricated of a corrosion resistant material.

Nominal installation torque. When card holders are used in cold plate applications, the nominal installation torque of each card holder shall be 4 to 6 inch-pounds (0.45 to 0.68 N-m).

Cold plate dimensions. The recommended cold plate slot width to accommodate the circuit card assembly with attached card holder is .210 inch (5.33 mm) plus the thickness of the printed board of the circuit card assembly (see A–A–59590).

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Inches	mm	Inches	mm	Inches	mm
.08	2.0	.220	5.59	.42	10.7
.180	4.57	.240	6.1		

NOTES:

1. Dimensions are in inches. Millimeters are given for information only.
2. Unless otherwise specified, tolerances are for ± .02 inch (0.5 mm) for two place decimals and ± .010 (0.25 mm) for three place decimals.
3. Dimension shall be .20 (5.1 mm) maximum on card holders without option “W” or “D” and .30 (7.6 mm) maximum on card holders with option “W” or “D”.
4. The across flats dimension for hex drive socket shall be .094 inch (2.38 mm) for mounting options “J”, “K”, “N”, “T”, “R”, “V”, and “S”. The across flats dimension for hex drive socket shall be a metric dimension of 2.5 mm (0.98 inch) across flats for mounting options “G” and “M”.

FIGURE 1. Relaxed and expanded dimensions.



FIGURE 2. Screw retaining self-locking nut (corrosion resistant) details.

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Finish. The finish options shall be as specified in A–A–59590. The finish options available for this specification sheet are as follows: “B” (black anodize), “C” (chemical film), “E” (electroless nickel), “H” (hard black anodize), and “P” (electrodeposited nickel).

Mounting. The mounting options shall be as specified in A–A–59590. The mounting options available for this specification sheet are as follows: “J” or “K” (rivet mounting holes), “N” (no mounting holes), “T” or “R” (tapped 0-80 UNF 2B holes), “V” or “S” (tapped 2-56 UNC 2B holes), or “G” or “M” (tapped M2.5 x.45 metric holes). Card holders using mounting option “J” or “K” may be shipped unassembled.

Rivet mounting holes. The holes used for rivet mounting shall be .066 to .073 inch (1.68 to 1.85 mm) diameter through holes, countersunk 100 degrees by .140 inch (3.56 mm) diameter with an access/clearance counterbore hole of .156 inch (3.96 mm) diameter by .130 to .140 inch (3.30 to 3.56 mm) deep.

Mounting hole locations (when required). Two mounting holes are required on card holders less than 5.30 inches (134.6 mm) in length. Three mounting holes are required on card holders 5.30 inches (134.6 mm) or greater in length. The third mounting hole shall be centered on the mounting body. See figure 1 and table I for mounting hole locations and spacing requirements.

Length, expanded, and relaxed dimensions. The length, expanded, and relaxed dimensions shall be as specified on figure 1. The length designator shall be as specified in A–A–59590 and the available lengths for this specification sheet are listed in table I.

TABLE I. Additional assembly dimensions (see figures 1 and 2). 1/

PIN length designator	Dimension “LENGTH” ± .02 (0.5)	Dimension “A” (maximum)		Dimension “B” ± .02 (0.5)	Dimension “C” ± .005 (0.13)	Dimension “D” ± .01 (0.3)
		Without options “W” or “D”	With options “W” or “D”			
18	1.8 (45.7)	2.10 (53.3)	2.20 (55.9)	.95 (24.1)	.40 (10.2)	.275 (6.99)
23	2.3 (58.4)	2.60 (66.0)	2.70 (68.6)	1.45 (36.8)	.70 (17.8)	.370 (9.39)
28	2.8 (71.1)	3.10 (78.7)	3.20 (81.3)	1.95 (49.5)	.90 (22.9)	.525 (13.3)
30	3.0 (76.2)	3.30 (83.8)	3.40 (86.4)	2.15 (54.6)	1.10 (27.9)	.525 (13.3)
33	3.3 (83.8)	3.60 (91.4)	3.70 (94.0)	2.45 (62.2)	1.40 (35.6)	.525 (13.3)
35	3.5 (88.9)	3.80 (96.5)	3.90 (99.1)	2.65 (67.3)	1.60 (40.6)	.525 (13.3)
38	3.8 (96.5)	4.10 (104.1)	4.20 (106.7)	2.95 (74.9)	1.90 (48.3)	.525 (13.3)
40	4.0 (101.6)	4.30 (109.2)	4.40 (111.8)	3.15 (80.0)	2.10 (53.3)	.525 (13.3)
43	4.3 (109.2)	4.60 (116.8)	4.70 (119.4)	3.45 (87.6)	2.40 (60.9)	.525 (13.3)
45	4.5 (114.x)	4.80 (121.9)	4.90 (124.5)	3.65 (92.7)	2.60 (66.0)	.525 (13.3)
48	4.8 (121.9)	5.10 (129.5)	5.20 (132.1)	3.95 (100.3)	2.90 (73.7)	.525 (13.3)
50	5.0 (127.0)	5.30 (134.6)	5.40 (137.2)	4.15 (105.4)	3.10 (78.7)	.525 (13.3)
53	5.3 (134.6)	5.60 (142.2)	5.70 (144.8)	4.45 (113.0)	3.40 (86.4)	.525 (13.3)
58	5.8 (147.3)	6.10 (154.9)	6.20 (157.5)	4.95 (125.7)	3.90 (99.1)	.525 (13.3)

1/ Dimensions are in inches. Millimeters, in parenthesis, are given for information only.

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Options. Card holders can have the following options. Card holders not requiring the following option shall leave the options position in the PIN blank.

Lockwasher and flat washer (see figure 3). The use of a lockwasher and flat washer located under the screw head will provide additional resistance to loosening from shock and vibration. Card holders requiring a lockwasher and flat washer option shall include a suffix “W” in the PIN (see classification and notes).

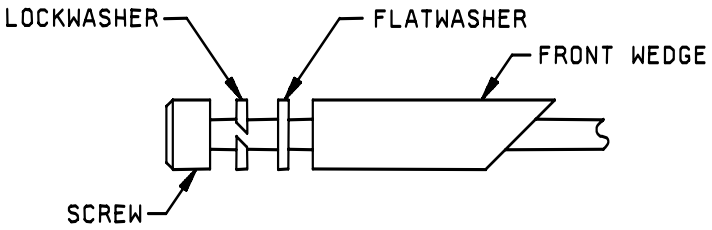


FIGURE 3. Lockwasher and flat washer details.

Screw self-locking element (see figure 4). The use of a screw self-locking element will provide prevailing torque for resistance to loosening from shock and vibration. Card holders requiring a screw self-locking element shall include a suffix “E” in the PIN (see classification and notes).



FIGURE 4. Screw self-locking element details.

Lockwasher, flat washer, and screw self-locking element. Card holders requiring a lockwasher, flat washer, and screw self-locking element option shall include a suffix “D” in the PIN (see classification and notes).

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

Commercial Item Description

A–A–59590 – Holder, Electrical Card, Wedge Retainers, 3 Piece, Screw Actuated Drive, General Requirements For.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111–5094.)

Ordering data. Ordering data is as specified in A–A–59590.

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Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of preparation and coordination of this CID specification sheet, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

Manufacturer CAGE	Manufacturer name and address	Manufacturer contact information
18915	Birtcher Pentair Electronic Packaging 14100 Danielson Street Poway, CA 92064–6898	Telephone: (858) 679–4550 Facsimile: (858) 679–4555 Electronic mail: infobirtcher@pentair-ep.com URL: www.birtcherproducts.com
52094	Calmark Corporation 4915 Walnut Grove Avenue San Gabriel, CA 91776–2099	Telephone: (626) 287–0451 Facsimile: (626) 287–7350 Electronic mail: sales@calmark.com URL: www.calmark.com

Part number supersession data. This CID specification sheet part numbers supersedes the following manufacturer's part numbers as shown in table II. The CID PINs listed in table II are only for length designator "28". See table III for CID PIN construction using other available lengths for this specification sheet. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE II. Commercial part number supersession data.

PIN designator AA59590/12	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/12	Vendor similar designator or type part number 1/ CAGE 52094
CJ28	41–6–LN	CJ28	240–2.80HR
CN28	41–6–LN–A	CN28	240–2.80R
CT28	41–6–LN–T	CT28	240–2.80T0R
CV28	41–6–LN–S	CV28	240–2.80T2R
CG28	41–6–LN–M	CG28	M240–2.80TM2.5R
CJ28E	41–6–LN–L	CJ28E	240–2.80HLR
CN28E	41–6–LN–A–L	CN28E	240–2.80LR
CT28E	41–6–LN–T–L	CT28E	240–2.80T0LR
CV28E	41–6–LN–S–L	CV28E	240–2.80T2LR
CG28E	41–6–LN–M–L	CG28E	M240–2.80TM2.5LR
CJ28W	41–6–LF–LN	CJ28W	W240–2.80HR
CN28W	41–6–LF–LN–A	CN28W	W240–2.80R
CT28W	41–6–LF–LN–T	CT28W	W240–2.80T0R
CV28W	41–6–LF–LN–S	CV28W	W240–2.80T2R
CG28W	41–6–LF–LN–M	CG28W	MW240–2.80TM2.5R

See footnote at end of table.

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TABLE II. Commercial part number supersession data – Continued.

PIN designator AA59590/12	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/12	Vendor similar designator or type part number 1/ CAGE 52094
CJ28D	41–6–LF–LN–L	CJ28D	W240–2.80HLR
CN28D	41–6–LF–LN–A–L	CN28D	W240–2.80LR
CT28D	41–6–LF–LN–T–L	CT28D	W240–2.80T0LR
CV28D	41–6–LF–LN–S–L	CV28D	W240–2.80T2LR
CG28D	41–6–LF–LN–M–L	CG28D	MW240–2.80TM2.5LR
BJ28	41–6–B–LN	BJ28	A240–2.80HR
BN28	41–6–B–LN–A	BN28	A240–2.80R
BT28	41–6–B–LN–T	BT28	A240–2.80T0R
BV28	41–6–B–LN–S	BV28	A240–2.80T2R
BG28	41–6–B–LN–M	BG28	MA240–2.80TM2.5R
BJ28E	41–6–B–LN–L	BJ28E	A240–2.80HLR
BN28E	41–6–B–LN–A–L	BN28E	A240–2.80LR
BT28E	41–6–B–LN–T–L	BT28E	A240–2.80T0LR
BV28E	41–6–B–LN–S–L	BV28E	A240–2.80T2LR
BG28E	41–6–B–LN–M–L	BG28E	MA240–2.80TM2.5LR
BJ28W	41–6–B–LF–LN	BJ28W	WA240–2.80HR
BN28W	41–6–B–LF–LN–A	BN28W	WA240–2.80R
BT28W	41–6–B–LF–LN–T	BT28W	WA240–2.80T0R
BV28W	41–6–B–LF–LN–S	BV28W	WA240–2.80T2R
BG28W	41–6–B–LF–LN–M	BG28W	MWA240–2.80TM2.5R
BJ28D	41–6–B–LF–LN–L	BJ28D	WA240–2.80HLR
BN28D	41–6–B–LF–LN–AL	BN28D	WA240–2.80LR
BT28D	41–6–B–LF–LN–TL	BT28D	WA240–2.80T0LR
BV28D	41–6–B–LF–LN–SL	BV28D	WA240–2.80T2LR
BG28D	41–6–B–LF–LN–ML	BG28D	MWA240–2.80TM2.5LR
HJ28	41–6–B3–LN	HJ28	HA240–2.80HR
HN28	41–6–B3–LN–A	HN28	HA240–2.80R
HT28	41–6–B3–LN–T	HT28	HA240–2.80T0R
HV28	41–6–B3–LN–S	HV28	HA240–2.80T2R
HG28	41–6–B3–LN–M	HG28	MHA240–2.80TM2.5R

See footnote at end of table.

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TABLE II. Commercial part number supersession data – Continued.

PIN designator AA59590/12	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/12	Vendor similar designator or type part number 1/ CAGE 52094
HJ28E	41–6–B3–LN–L	HJ28E	HA240–2.80HLR
HN28E	41–6–B3–LN–A–L	HN28E	HA240–2.80LR
HT28E	41–6–B3–LN–T–L	HT28E	HA240–2.80T0LR
HV28E	41–6–B3–LN–S–L	HV28E	HA240–2.80T2LR
HG28E	41–6–B3–LN–M–L	HG28E	MHA240–2.80TM2.5LR
HJ28W	41–6–B3–LN–LF	HJ28W	WHA240–2.80HR
HN28W	41–6–B3–LF–LN–A	HN28W	WHA240–2.80R
HT28W	41–6–B3–LF–LN–T	HT28W	WHA240–2.80T0R
HV28W	41–6–B3–LF–LN–S	HV28W	WHA240–2.80T2R
HG28W	41–6–B3–LF–LN–M	HG28W	MWHA240–2.80TM2.5R
HJ28D	41–6–B3–LF–LN–L	HJ28D	WHA240–2.80HLR
HN28D	41–6–B3–LF–LNAL	HN28D	WHA240–2.80LR
HT28D	41–6–B3–LF–LNTL	HT28D	WHA240–2.80T0LR
HV28D	41–6–B3–LF–LNSL	HV28D	WHA240–2.80T2LR
HG28D	41–6–B3–LF–LNML	HG28D	MWHA240–2.80TM2.5LR
PJ28	41–6–N–LN	EJ28	EN240–2.80HR
PN28	41–6–N–LN–A	EN28	EN240–2.80R
PT28	41–6–N–LN–T	ET28	EN240–2.80T0R
PV28	41–6–N–LN–S	EV28	EN240–2.80T2R
PG28	41–6–N–LN–M	EG28	MEN240–2.80TM2.5R
PJ28E	41–6–N–LN–L	EJ28E	EN240–2.80HLR
PN28E	41–6–N–LN–A–L	EN28E	EN240–2.80LR
PT28E	41–6–N–LN–T–L	ET28E	EN240–2.80T0LR
PV28E	41–6–N–LN–S–L	EV28E	EN240–2.80T2LR
PG28E	41–6–N–LN–M–L	EG28E	MEN240–2.80TM2.5LR
PJ28W	41–6–N–LN–LF	EJ28W	WEN240–2.80HR
PN28W	41–6–N–LF–LN–A	EN28W	WEN240–2.80R
PT28W	41–6–N–LF–LN–T	ET28W	WEN240–2.80T0R
PV28W	41–6–N–LF–LN–S	EV28W	WEN240–2.80T2R
PG28W	41–6–N–LF–LN–M	EG28W	MWEN240–2.80TM2.5R

See footnote at end of table.

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TABLE II. Commercial part number supersession data – Continued.

PIN designator AA59590/12	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/12	Vendor similar designator or type part number 1/ CAGE 52094
PJ28D	41–6–N–LF–LN–L	EJ28D	WEN240–2.80HLR
PN28D	41–6–N–LF–LNAL	EN28D	WEN240–2.80LR
PT28D	41–6–N–LF–LNTL	ET28D	WEN240–2.80T0LR
PV28D	41–6–N–LF–LNSL	EV28D	WEN240–2.80T2LR
PG28D	41–6–N–LF–LNML	EG28D	MWEN240–2.80TM2.5LR

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.

TABLE III. Example of PIN with available length designators.

PIN designator AA59590/12	Vendor similar designator or type part number 1/ 2/ CAGE 18915	PIN designator AA59590/12	Vendor similar designator or type part number 1/ 2/ CAGE 52094
CT18	41–4–LN	CT18	240–1191T0R
CT23	41–5–LN–T	CT23	240–2.30T0R
CT28	41–6–LN–T	CT28	240–2.80T0R
CT30	3/	CT30	240–3.00T0R
CT33	41–7–LN–T	CT33	240–3.30T0R
CT35	3/	CT35	240–3.50T0R
CT38	41–8–LN–T	CT38	240–3.80T0R
CT40	3/	CT40	240–4.00T0R
CT43	41–9–LN–T	CT43	240–4.30T0R
CT45	3/	CT45	240–4.50T0R
CT48	41–10–LN–T	CT48	240–4.80T0R
CT50	3/	CT50	240–5.00T0R
CR53	41–11–LN–T	CR53	240–5.30ET0R
CR58	41–12–LN–T	CR58	240–5.80ET0R

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.
2/ Other lengths are available on request.
3/ Consult manufacturer for part number.

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MILITARY INTERESTS:

Custodians:
Army – CR
Navy – EC
Air Force – 11
DLA – CC

Review activity:
Air Force – 99

CIVIL AGENCY COORDINATING ACTIVITY:

GSA – FSS

Preparing Activity:
DLA – CC

Project 5998–2005–011

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